

Demo: Loading Macros via include shortcode

This page demonstrates the [include shortcode](#) strategy for loading macros into a Quarto document.

The `{{< include macros.qmd >}}` shortcode at the top of this file embeds the entire contents of `macros.qmd` inline at that point. Since `macros.qmd` contains only raw LaTeX macro definitions (`\def`, `\providecommand`, etc.), Quarto passes them through as raw LaTeX for PDF output (where they are processed inline in the document body as LaTeX commands) and as math for HTML/RevealJS output (where they are processed by MathJax).

Note: This demo uses `macros.qmd` directly (same directory). When using this repository as a submodule, use `macros/macros.qmd` instead.

Example math

The table below uses macros defined with `\def` in `macros.qmd`. These macros work across all output formats:

Description	Code	Output
Greek: alpha	<code> \$\a\$ </code>	α
Greek: beta	<code> \$\b\$ </code>	β
Greek: gamma	<code> \$\g\$ </code>	γ
Greek: lambda	<code> \$\l\$ </code>	λ
Greek: sigma	<code> \$\s\$ </code>	σ
Log-likelihood	<code> \$\llik\$ </code>	ℓ
Independent	<code> \$X \ind Y\$ </code>	$X \perp\!\!\!\perp Y$
IID distributed	<code> \$X_i \siid \Normal(\m, \ss)\$ </code>	$X_i \sim_{\text{iid}} N(\mu, \sigma^2)$
Vector: mu	<code> \$\vm\$ </code>	$\tilde{\mu}$
Hat beta	<code> \$\hb\$ </code>	$\hat{\beta}$
Sigma-algebra	<code> \$\salg\$ </code>	$\mathcal{S}$
Borel sigma-algebra	<code> \$\borel\$ </code>	$\mathcal{B}$

Color macros

`macros.qmd` defines a one-argument macro for each of the 19 [LaTeX base colors](#) (`\red`, `\blue`, `\green`, `\orange`, ...), using the `\color` switch form (`{\color{red}{#1}}`), which is supported by MathJax (HTML/RevealJS), lualatex (PDF), and pandoc's texmath parser (docx):

Description	Code	Output
Red	<code> \$\red{x}\$ </code>	x
Blue	<code> \$\blue{y}\$ </code>	y
Green	<code> \$\green{z}\$ </code>	z
Orange	<code> \$\orange{\mu}\$ </code>	μ
Combined	<code> \$\red{x} + \blue{y} = \green{z}\$ </code>	$x + y = z$
Nested macros	<code> \$X_i \sim_{\red{\ind}} \blue{\Normal}(\orange{\mu_i}, \green{\ss})\$ </code>	$X_i \sim_{\textcolor{red}{\texttt{ind}}} \textcolor{blue}{N}(\textcolor{orange}{\mu_i}, \textcolor{green}{\sigma^2})$

Note (docx): pandoc converts these equations to native Word math without warnings, but its docx writer drops the color styling, so in Word the equations render correctly in the default text color.

`\providecommand` behavior in this shortcode demo

The macro `\floor` in `macros.qmd` is defined with `\providecommand`. Compare this with a `\def` macro:

Macro type	Code	Output
<code>\providecommand macro (\floor)</code>	<code> \$\floor{1.2}\$ </code>	$\lfloor 1.2 \rfloor$
<code>\def macro (\a)</code>	<code> \$\a\$ </code>	α

In this shortcode demo, both macros render in HTML/RevealJS. So in this setup, `\providecommand` from the included source is being handled correctly.

Note (RevealJS blank slide): Content placed before the first `##` heading in RevealJS output becomes a blank title slide. Since the include shortcode is at the top of this file, RevealJS may show a blank slide before the first section. This is a [known limitation](#) of the shortcode approach for RevealJS. For RevealJS without a blank slide, use the [include-in-header strategy](#) instead.